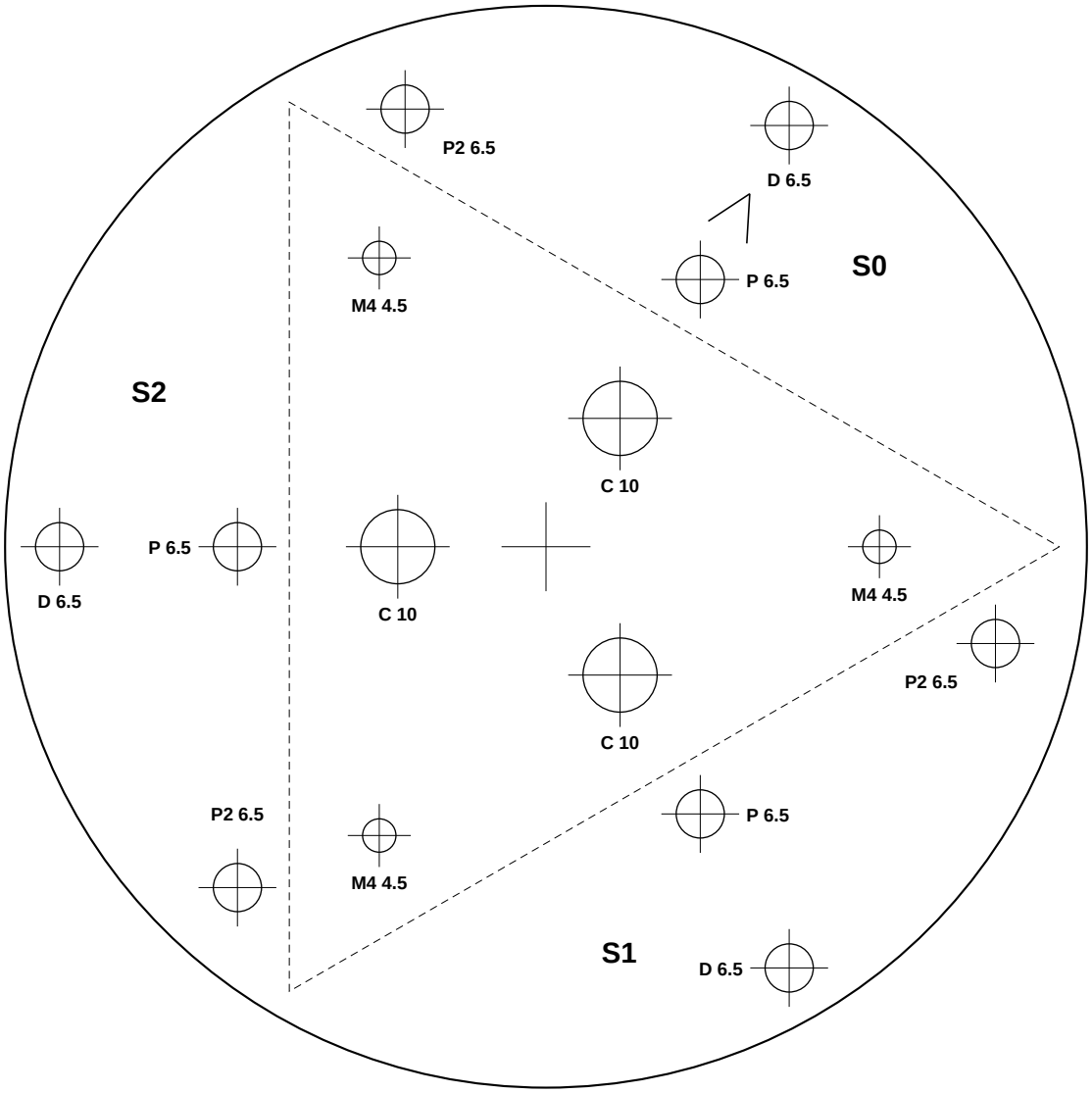
Shield deck drill template - 146 mm disc - PRINT AT 100% (Actual size)

Viewed from the TOP (electronics side). Tape to the board, centre-punch each crosshair, then drill.



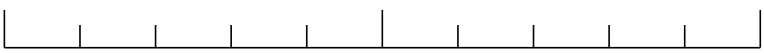
Solid circle = cut line (146 mm). Dashed triangle = where the 120 mm back panels stand, copper side out.

D 6.5 mm = rubber duck / monopole (31 mm in front of face) P / P2 6.5 mm = patch (7 mm in front of face; P2 14 mm from the face end)

C 10 mm = dipole coax pass-through (inside core) M4 4.5 mm = bolts to the electronics plate (inside core)

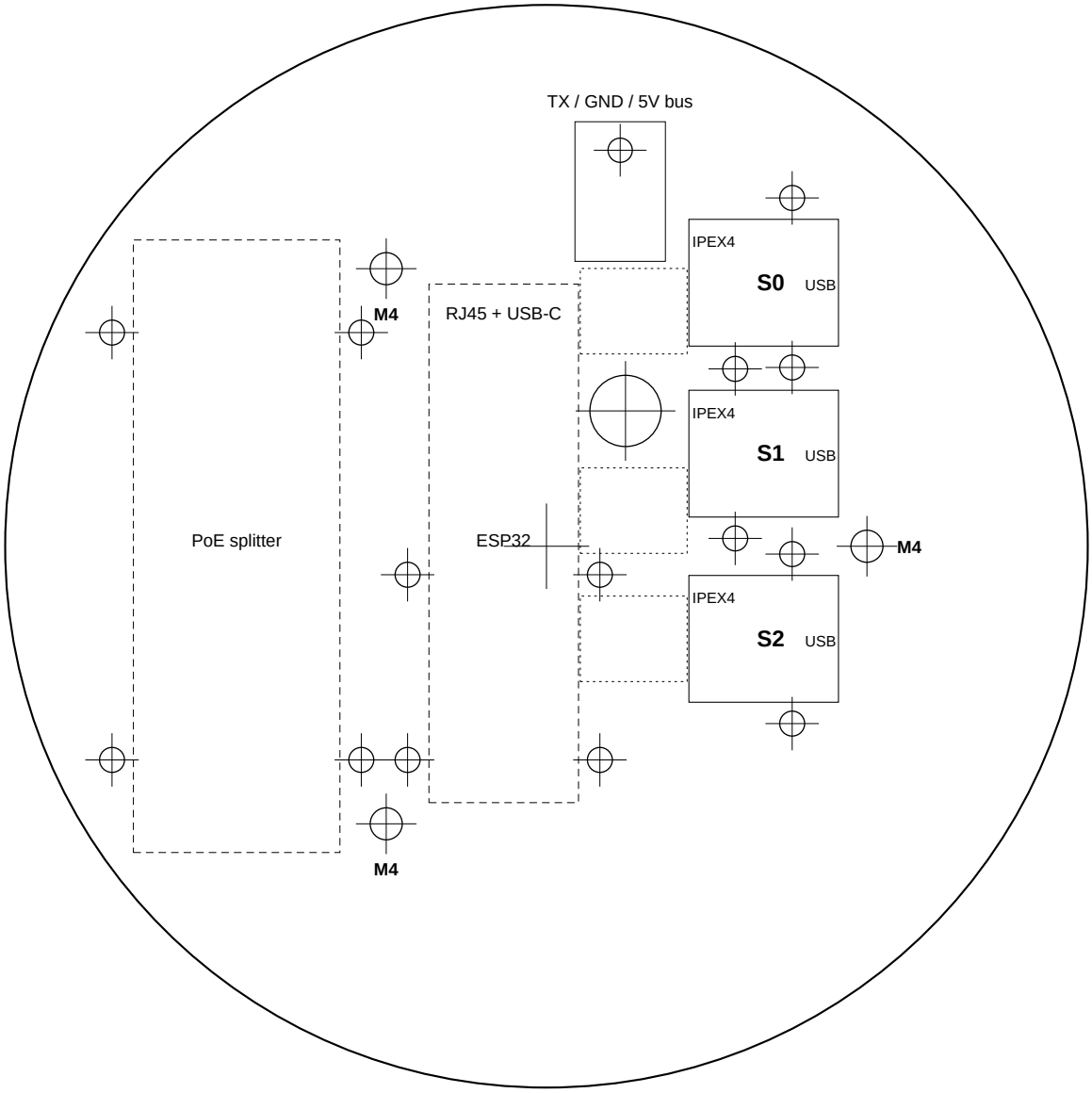
Arrow points to Sector 0. Holes are symmetric, so the template also works flipped (labels read mirrored).

Drawn the same way round as the plate (page 2): plate 12 o'clock up, S0 arrow 30 deg clockwise of it.

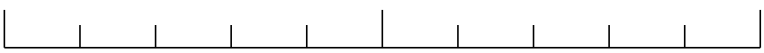


Electronics plate template - 152 mm disc - PRINT AT 100% (Actual size)

Viewed from the TOP (parts side). Gateway left, scanners right, one pigtail hole and the JST pairs between.

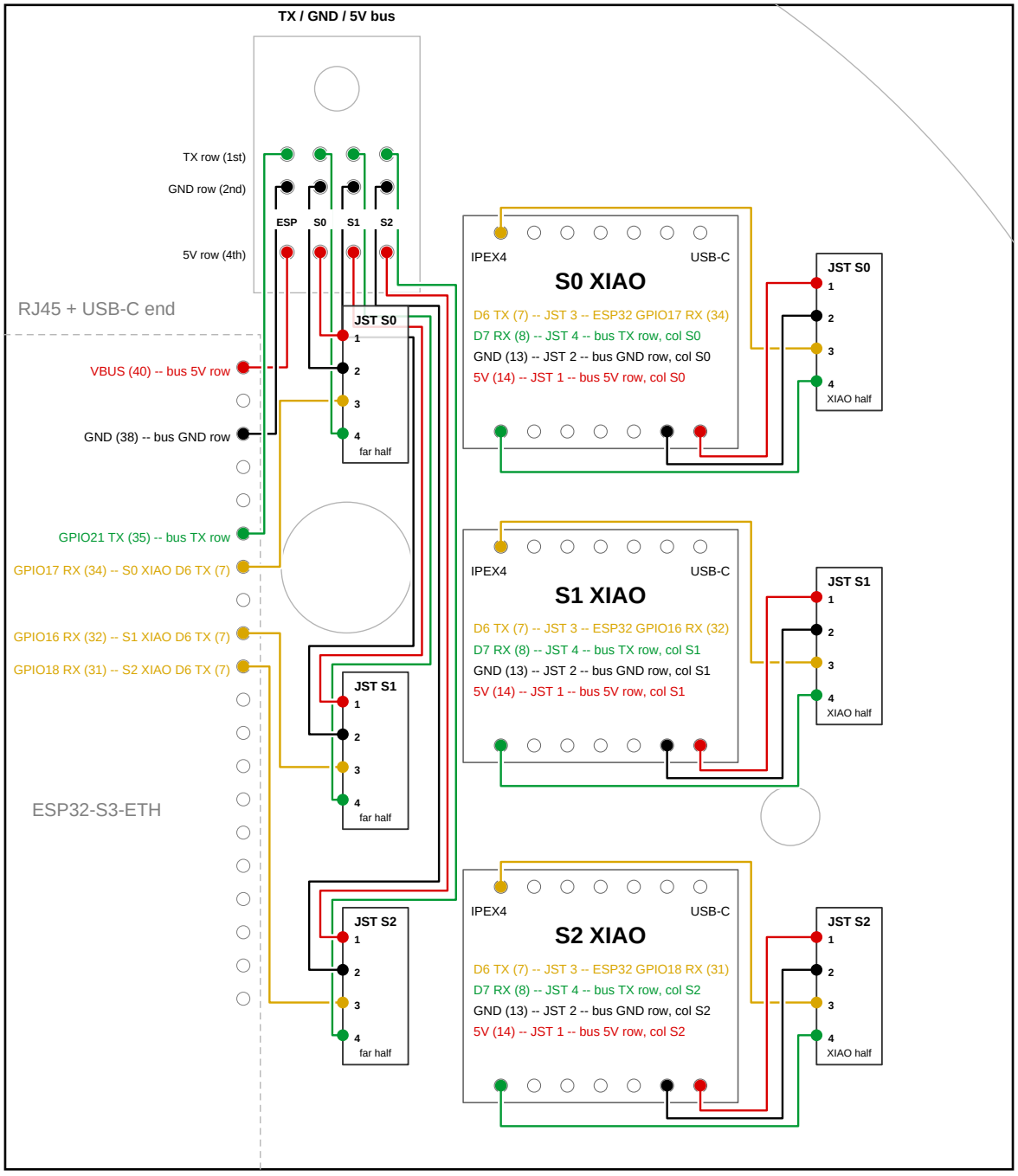


10 mm: all three pigtails (MHF4 ends threaded up from below), level with S1's IPEX4 connector -
assumed 6 mm above the XIAO centre line: check on the board. 3.4 mm (M3): bus board.
3 x 4.5 mm (M4, labelled): bolts to the deck, plate 12 o'clock 30 deg counter-clockwise of the deck's S0 arrow, ~25 mm up.
Small holes: 3.5 mm zip-tie pairs: one tie across each XIAO (USB-C toward the rim), two each across
splitter and ESP32. Coax strain relief: grommet in the pigtail hole, or a dab of RTV once the coax are placed.
Dotted: mated JST pairs (unplug to reflash); far side D6 to its ESP32 RX pin, 3 wires to the bus.
Dashed: ESP32-S3-ETH 72.8 x 21 mm; splitter from a photo - lay it on the printout, tie holes 3 mm outside its edges.



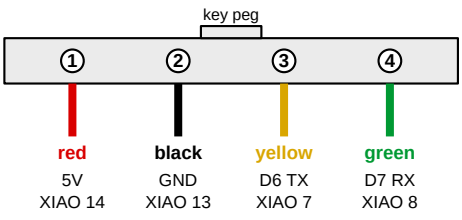
Electronics plate wiring - same layout as the drill template

Top view of the radio side, 2x (not for drilling). Pin numbers in brackets. Pad positions approximate.
JST halves drawn apart: XIAO half right of each radio, far half in the strip; same name = they plug together.



D6 -> its own ESP RX (the pin is the sector): S0 GPIO17 (34), S1 GPIO16 (32), S2 GPIO18 (31).
ESP TX GPIO21 (35) -> bus TX row -> every D7. Feed: TX (35) -> TX row, VBUS (40) -> 5V row, GND (38) -> GND row.
Bus: hole end up, rows bridged, 3rd row spare. ESP32 RJ45 side up, RJ45 / USB-C end north: VBUS (40) top right pin.

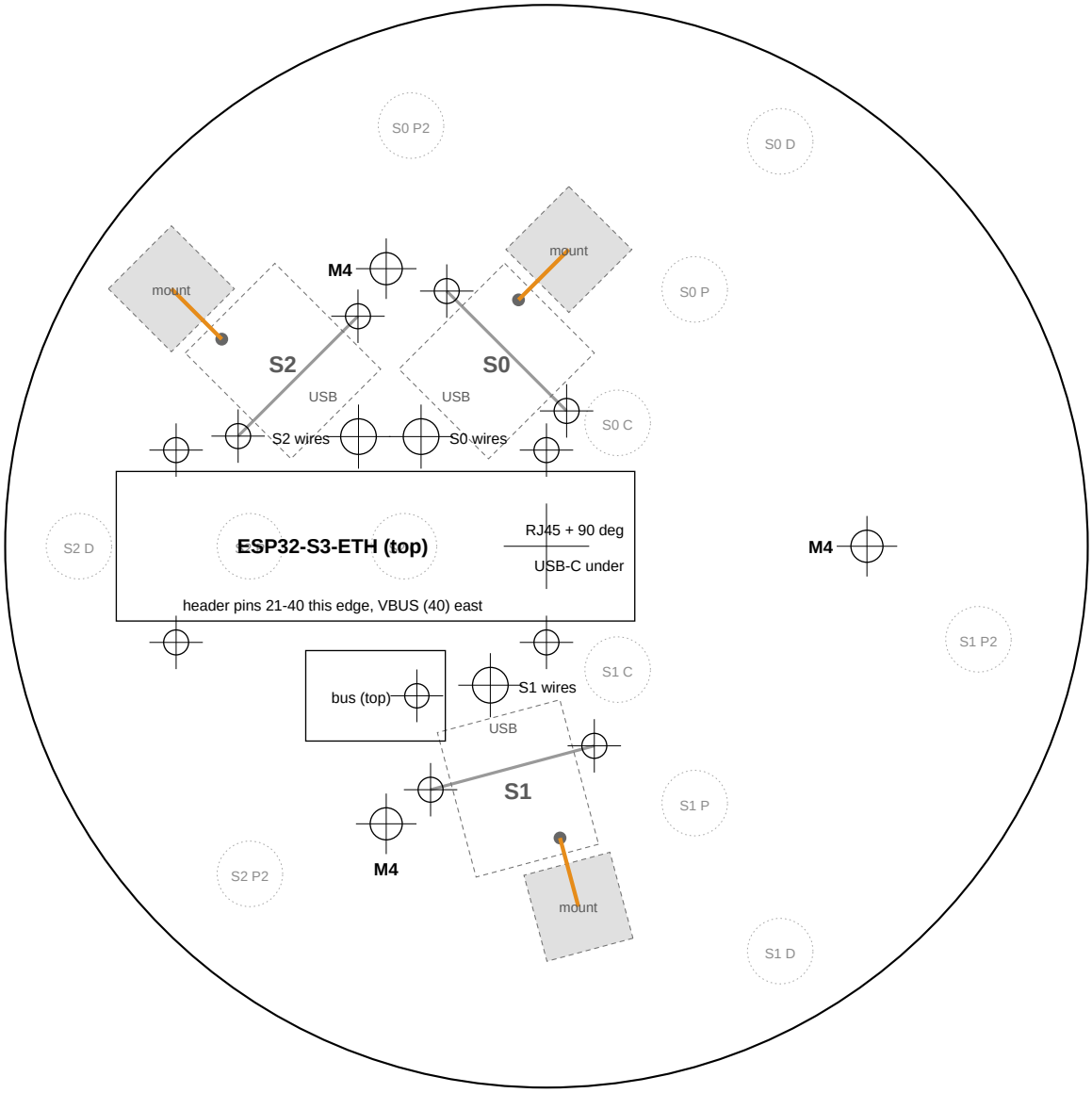
JST head (either half), crimp side, key peg up:



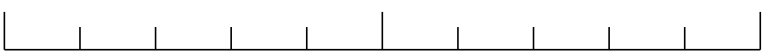
Pins left to right: 1 red 5V, 2 black GND, 3 yellow D6 TX, 4 green D7 RX.
Male and female halves carry the same colours in the same order on the crimp side, so the XIAO tail and the far tail match colour for colour through the joint.
XIAO tail: red -> 5V (14), black -> GND (13), yellow -> D6 (7), green -> D7 (8).
Far tail: red -> bus 5V row, black -> bus GND row, green -> bus TX row, yellow -> that sector's ESP RX pin (labels on the drawing). Female half on the far tail keeps 5V covered.

ALT electronics plate (split) - 152 mm disc - PRINT AT 100% (Actual size)

Viewed from the TOP. Radios on the UNDERSIDE (grey, seen through the plate); ESP32 + bus on top. No pigtail hole.



Drill: 3 x 4.5 mm (M4) bolts to the deck, same as page 2; 3.4 mm (M3) bus board; 3 x 5 mm harness holes (chamfered): each XIAO's 4-wire tail up to its JST pair on top; 3.5 mm pairs: one tie across each XIAO, two across the ESP32.
Underside (grey): XIAOs on VHB + a zip tie (grey bar), IPEX4 dot -> orange flat coax run -> HS-100 tie mount, which clamps it; bend only after the mount, down to its deck jack. Dotted circles: deck jacks below, not drilled.
Top: ESP32 RJ45 / USB-C end east, near the centre. A 90 deg adapter turns the RJ45 up; patch cable and the USB-C power lead run through the 4" box's 1" hole to the PoE splitter. USB-C is under the board: slim plug.
Spacers 25-30 mm: the radios and the bulkhead backs share that gap.

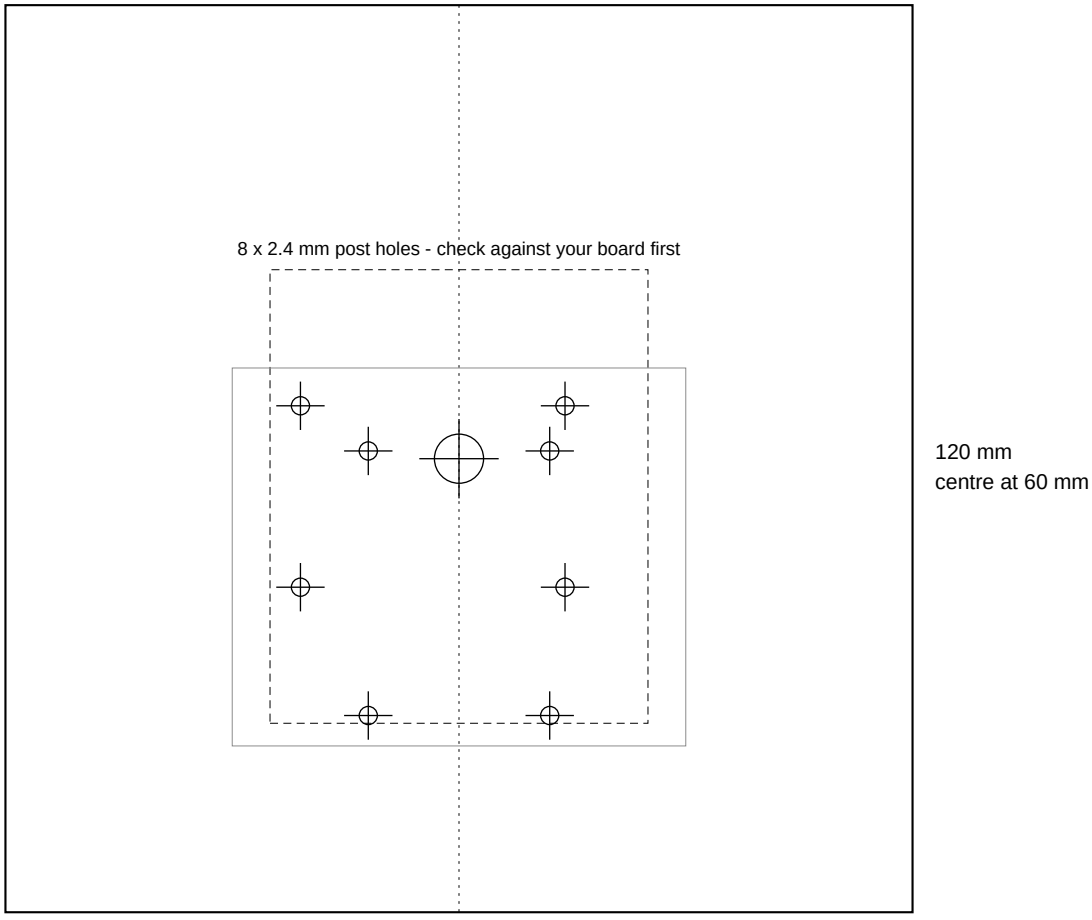


Back panel (reflector face) template - 120 x 120 mm - PRINT AT 100%

Make 3. Viewed from the FRONT (copper side, facing the antenna). Top edge goes against the shield deck.

0 100 mm - check with a ruler before drilling

TOP - against shield deck



A (dashed): bar vertical, notch right, board 35-95 mm down - stalk reaches the P2 deck jack

C (grey): bar horizontal, notch up, board 48-98 mm down

Centre 6.5 mm hole: DIY dipole RP-SMA only

Rubber duck / monopole / dipole sit 31 mm in front of this face (on the D hole).

Panel board 50 x 60 mm: 7 mm off this face on nylon standoffs. A = stalk on the P2 deck jack; C = coax may need extending.

Corners meet at 120 deg to form the triangle; solder or tape the seams. Holes under 10 mm don't affect the reflector.

0 100 mm - check with a ruler before drilling